Year 10 Science
Physical Sciences
Worksheet - Displacement (Extension)

RESULTANT OF TWO (OR MORE) DISPLACEMENTS

When a body is subjected to two (or more) displacements the final direct distance (the displacement) from the starting position is *NOT* equal to the total distance travelled (unless the individual displacements are in the same direction).

To determine the exact location (distance and direction from the starting position) of a body, it is necessary to determine the *RESULTANT* displacement.

A resultant of two (or more) vectors may be defined as the single vector which on its own will produce the same effect as the two (or more) vectors combined.

DETERMINATION OF RESULTANTS

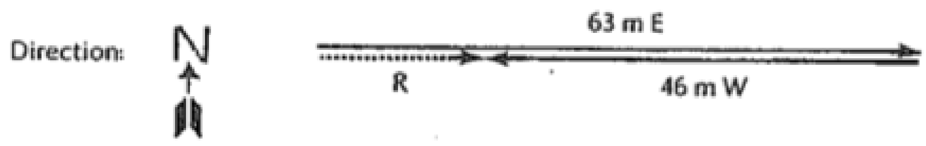
To determine the resultant:

1. Place the vectors head to tail (in any order).
2. Draw in the resultant which is from the tail of the first vector to the head of the last.
3. If the vectors have been drawn to scale then the magnitude of the resultant and the direction can be measured — otherwise both can be calculated.

This method is often referred to as the *"head to tail"* or *"triangle"* method.

TYPE EXAMPLE 5: What is the resultant displacement of a displacement of 63 m east and a displacement of 46 m west?

Sketch vectors head to tail (approximately to scale) and the resultant displacement.



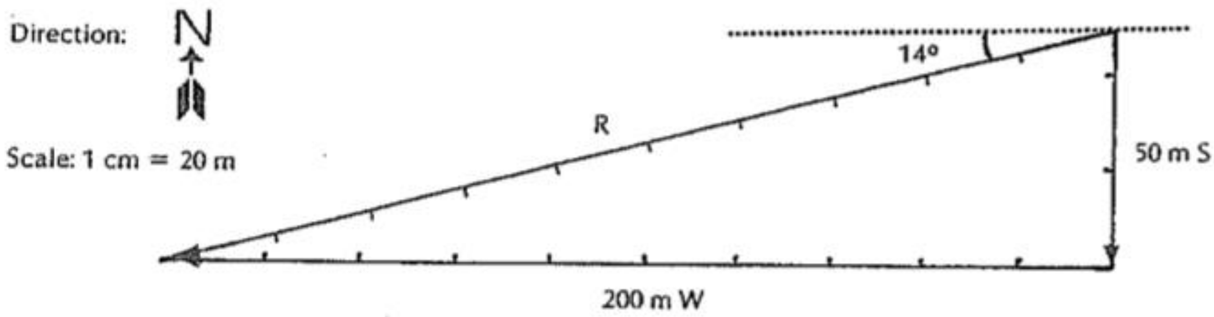
For displacements in one straight line, measurement (i.e. graphical solution) is unnecessary as the calculation of the magnitude of the resultant is obvious.

$$\text{resultant, R} = (63\text{ m}) - (46\text{ m}) = 17\text{ m}$$

i.e. the resultant displacement is 17 m east.

TYPE EXAMPLE 6: Determine the resultant displacement if a person walks 50 m south to a corner and then 200 m west to go to the local store.

A. GRAPHICAL SOLUTION

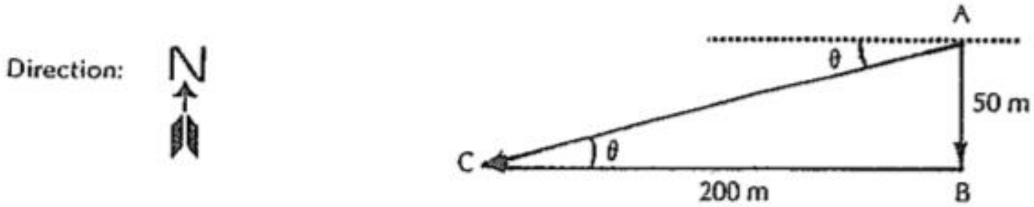


$$\text{length of R} = 10.3 \text{ cm}$$
$$\therefore \text{magnitude of R} = (10.3 \times 20) \text{ m} = 206 \text{ m}$$

i.e. the resultant displacement is 206 m W 14° S

B. CALCULATION SOLUTION (right angled triangles only).

1. Sketch the vector triangle (approximately to scale).



2. Using Pythagorus' Theorem calculate a value for the magnitude of the resultant, AC.

AC = resultant
= ?
AB = 50 m S
BC = 200 m W

$$\begin{aligned} AC^2 &= AB^2 + BC^2 \\ &= (50)^2 + (200)^2 \\ &= 2\,500 + 40\,000 \\ &= 42\,500 \\ \therefore AC &= 206.2 \text{ m} \end{aligned}$$

3. Using trigonometric ratios determine the direction of the resultant.

$$\begin{aligned} \tan\theta &= \frac{AB}{BC} \\ &= \frac{50}{200} \\ &= 0.25 \\ \therefore \theta &= 14° \end{aligned}$$

i.e. the resultant displacement is 206.2 m W 14° S.

SET 5: RESULTANT DISPLACEMENT III

Using Pythagorus' Theorem and trigonometric ratios determine the resultant displacement for each of the following:

1. 300 km N, 300 km W.
2. A climb 6 m out of a well and 8 m away from it.
3. 500 m E, 1 200 m S.
4. 17.32 m horizontally then 10 m vertically upwards.
5. 1.2 km NW, 1.6 km NE.

SET 5

1. 424 km NW **2.** 10 m 53° to the vertical **3.** 1 300 m E 67° S
4. 20 m 30° to the horizontal **5.** 2 km N 8° E